

Experiment 9: Rotation of an Image

Aim

To rotate an image in clockwise and counter-clockwise directions using OpenCV.

Procedure

1. Import OpenCV.
2. Read the input image.
3. Find the center of the image.
4. Create a rotation matrix.
5. Rotate the image using `cv2.warpAffine()`.
6. Display the rotated image.

Code:

```
import cv2

img = cv2.imread("image.jpg")

(h, w) = img.shape[:2]

center = (w // 2, h // 2)

matrix = cv2.getRotationMatrix2D(center, 90, 1.0)

rotated = cv2.warpAffine(img, matrix, (w, h))

cv2.imshow("Rotated Image", rotated)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

Output:



Result: The image was successfully rotated.